However, the hot water tank is likely to need to be reheated before the homeowner comes home in the evening. The heat pump is not aware that the property also has a solar PV inverter which is also an ESA that is communicating with the EMS.

The EMS first requests the Forecast data from each of its registered ESAs. It determines that the heat pump has a power profile suggesting it needs to heat hot water around 6pm. The solar PV inverter has forecast that it will generate 3.6kW of power during the middle of the day and into the afternoon before the sun goes down.

The EMS can then optimize the home considering other non-ESA loads and can ask the heat pump to heat the hot water around 3pm when it has forecast that excess solar power will be available.

It does this by sending a ModifyForecastRequest to the heat pump and asks the heat pump to expect to run at a lower power consumption (within the solar excess power) which requires the heat pump to run for a longer duration to achieve its required energy demand.

9.2.4.7. ConstraintBasedAdjustment Feature

ESAs which support the Constraint-Based Adjustment feature allow an EMS to inform the ESA of periods during which power usage should be modified (for example when the EMS has been made aware that the grid supplier has requested reduced energy usage due to overall peak grid demand) and may cause the ESA to modify the intended power profile has previously suggested it would use.

EVSE example: An EVSE scheduling system may have determined that the vehicle would be charged starting at a moderate rate at 1am, so that it has enough charge by the time it is needed later that morning.

However, the DSR service provider has informed the EMS that due to high forecast winds it is now forecast that there will be very cheap energy available from wind generation between 2am and 3am.

The EMS first requests the Forecast data from each of its registered ESAs. It determines that the EVSE has a power profile suggesting it plans to start charging the vehicle at 1am.

The EMS can then try to reduce the cost of charging the EV by informing the EVSE of the desire to increase the charging between scheduled times.

It does this by sending a RequestConstraintBasedForecast to the EVSE and asks it to run at a higher NominalPower consumption during the constraint period, which may require it to decrease its charge rate outside the constraint period to achieve its required energy demand.

9.2.5. Dependencies

This cluster does not report electrical power and electrical energy. Devices that use this cluster SHALL also support the Electrical Power Measurement and optionally support the Electrical Energy Measurement cluster to allow an energy management system to perform its role. See Device Type library for device specific details.

